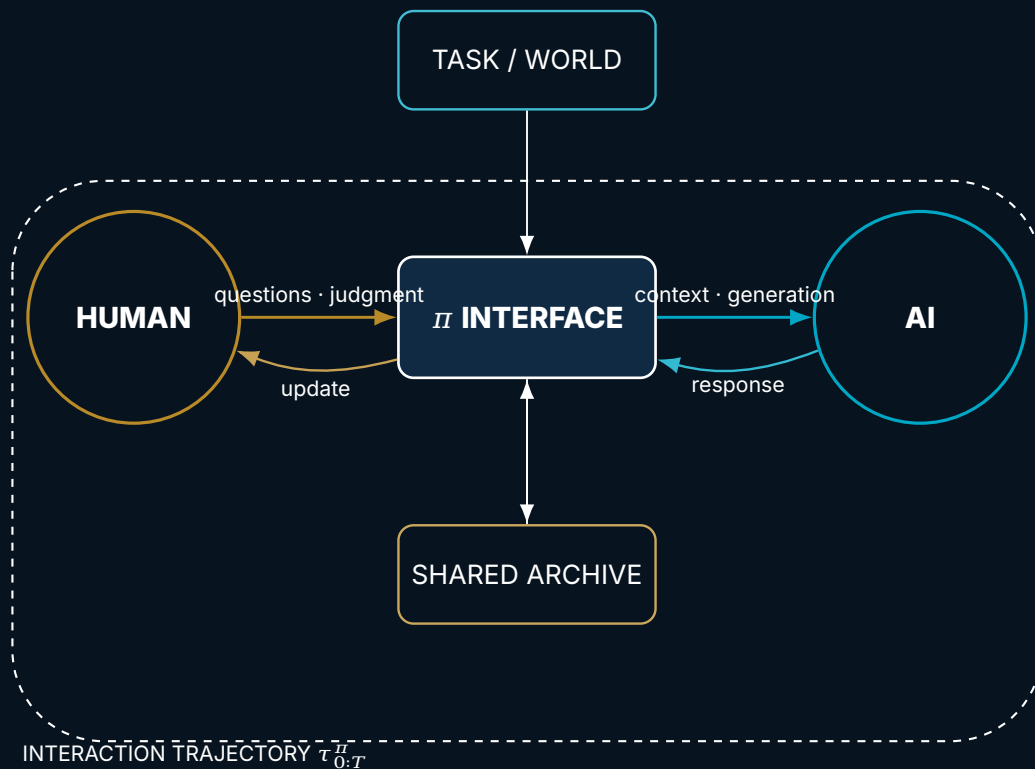


INTERFACE INTELLIGENCE

A Dynamical-Systems Research Program for Capability, Sovereignty, and Relational Cognition in Human–AI Systems



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Abstract

Hybrid intelligence is usually framed as a combination of human and machine capabilities. We argue that this framing is necessary but incomplete. In sustained human–AI collaboration, the scientifically relevant unit of analysis may often be neither the human nor the model in isolation, but the protocol-mediated interaction trajectory that couples them through dialogue, memory, tools, provenance, correction, and task structure. We call the measurable system-level properties of this trajectory interface intelligence. The term is deliberately operational rather than ontological: it does not posit a new conscious entity, nor does it imply that cognition literally resides in an interface. It identifies a research target—performance-relevant organization that appears only when human, model, and interaction history are treated as a coupled dynamical system.

The proposal is motivated by an empirical tension. Hybrid-intelligence research aims at outcomes unattainable by either humans or machines alone, yet a preregistered meta-analysis of 106 experiments found that human–AI combinations, on average, underperformed the better isolated component. This suggests that simply placing a human and an AI in the same loop is not sufficient for synergy. We therefore distinguish component capability from relational capability, and propose that human–AI systems be evaluated along a multidimensional interface signature: complementary task performance, synergistic information, bidirectional predictive coupling, path dependence, capability residue after tool withdrawal, opposition capacity, provenance fidelity, and transfer across model substitution.

We formalize the human–AI pair as a history-dependent coupled system whose transition law is altered by the interaction protocol. We then specify a six-arm experimental program comparing human-only, model-only, one-shot consultation, sustained dialogue, provenance-augmented dialogue, and model-swapped continuation. A physics-oriented benchmark based on mechanistic inference from synthetic dynamical systems provides ground truth while preserving room for scientific reasoning. The framework yields falsifiable predictions and a sovereignty vector that operationalizes exit, contestability, portability, traceability, and human authority without collapsing them into a single moral score. The central claim is modest but consequential: if interaction structure explains reliable variance that isolated component benchmarks cannot, then hybrid intelligence should be studied as an organized trajectory, not merely as the sum of its parts.

Claim boundary. Interface intelligence is an operational construct for measurable system-level properties of human–AI interaction trajectories. It does not assert a third consciousness, AI phenomenology, metaphysical fusion, or a new physical law.

Core thesis. The useful unit of analysis may often be neither the human nor the model in isolation, but the structured interaction trajectory through which they modify one another’s available state, hypotheses, errors, memory, and capability.

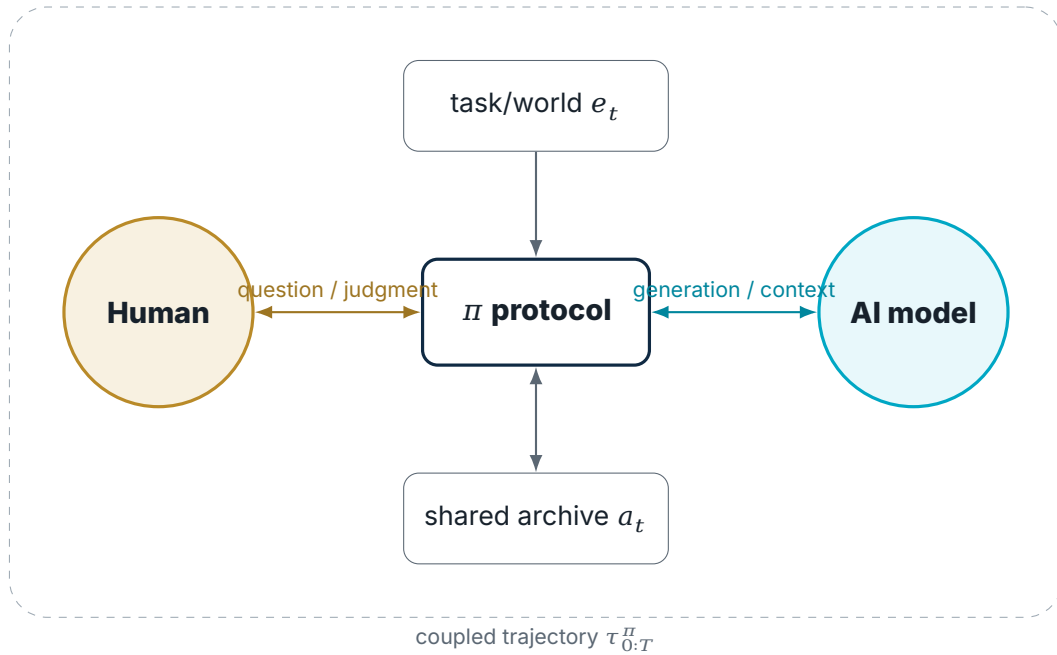
Interface signature at a glance

$$\mathbf{I}(\pi) = [C_{\text{perf}}, C_{\text{info}}, B_{\text{coup}}, P_{\text{path}}, R_{\text{cap}}, O_{\text{opp}}, V_{\text{prov}}, T_{\text{swap}}]$$

performance · information · coupling · path dependence · learning residue · opposition ·

provenance · portability

System map



Operational definition. Interface intelligence is the set of empirically measurable, task-relevant system properties whose explanatory or predictive value depends on the protocol-mediated interaction trajectory and cannot be adequately recovered from isolated human and model performance alone.

The diagram is intentionally agnostic about where cognition “really resides.” Its purpose is experimental: the protocol Π is a manipulable coupling architecture; the archive is an externalized state variable; the task/world supplies perturbations and ground truth; and the interaction trajectory is the object over which path dependence, learning, provenance, and transfer can be measured.

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1 Introduction: the missing object in human–AI research

The modern hybrid-intelligence literature already rejects a simple replacement narrative. Dellermann et al. define hybrid intelligence in terms of collective performance and reciprocal learning between heterogeneous intelligences, while Akata et al. define it as a combination of human and machine intelligence that augments human capability and achieves goals unreachable by either alone [1,2]. More recently, Natali has argued that emergence offers a unifying lens for hybrid intelligence: system-level properties can arise from organized couplings among humans, artefacts, and practices, and should therefore be evaluated at the level of the organized system rather than by component benchmarks alone [3].

The empirical record, however, makes the problem sharper. Vaccaro, Almaatouq, and Malone’s preregistered meta-analysis of 106 human-participant experiments and 370 effect sizes found that human–AI combinations, on average, performed significantly worse than the better of the human or AI alone, with substantial heterogeneity across task types [4]. Hemmer et al. similarly emphasize that complementary team performance—performance unattainable by either component alone—has been difficult to realize, and identify information and capability asymmetries as key sources of complementarity [5]. The implication is important: co-presence is not coupling, and coupling is not automatically synergy.

This paper proposes that a large part of the missing explanatory variable is the interaction architecture itself. A human and an AI model do not simply contribute two static capability vectors to a task. They generate a sequence of state changes through prompts, objections, revisions, tool calls, external memory, confidence updates, error corrections, and shifts in problem representation. The same human and the same model can therefore produce radically different outcomes under different interaction protocols. If those protocol-dependent differences are systematic, predictive, and experimentally manipulable, then they constitute a legitimate scientific object.

We call that object interface intelligence: the task-relevant system-level organization attributable to the history-dependent coupling among a human, an AI model, and the mediating interface. The phrase is not intended as a claim that an interface is conscious, possesses a mind, or forms a metaphysical “third entity.” It is an operational label for properties that disappear when the components are studied only in isolation.

2 Intellectual lineage: from distributed cognition to engineered hybrid systems

The proposal sits at the intersection of several traditions. Clark and Chalmers’ extended-mind thesis argued for active externalism: under appropriate coupling, environmental resources can participate constitutively in cognitive processes rather than merely supplying inputs [6]. Hutchins’ distributed cognition moved the unit of cognitive analysis from the isolated individual to culturally organized systems of people and artefacts, showing that system-level computational properties can differ from those of any individual participant [7]. Cognitive Systems Engineering and the joint-cognitive-systems tradition similarly treat human–technology ensembles as adaptive systems whose meaningful behavior cannot always be understood by decomposing them into independent parts [8].

Contemporary hybrid intelligence adds an engineering objective: not merely describing distributed cognition, but deliberately arranging roles, protocols, task dependencies, feedback, responsibility, and explanation so that human–AI systems outperform their components while remaining governable [2,3]. This difference matters. The present proposal therefore treats the interface not as a passive channel, but as a controllable coupling architecture.

The concept also connects to collective intelligence. Human groups can display system-level performance structure that is not trivially identical to the ability of their strongest member [9]. Human–AI systems add a deeper asymmetry: human and machine components differ not only in skill level but in representational form, memory constraints, failure modes, access to external information, temporal dynamics, and authority. The resulting ensemble is heterogeneous by construction.

3 Core proposition: interface intelligence is a trajectory property

Consider a task unfolding over discrete interaction steps $t = 0, 1, \dots, T$. Let h_t denote a representation of the human’s task-relevant state at step t : beliefs, hypotheses, confidence, working representation, and available knowledge. Let c_t denote the model-context state: the conversation history, retrieved material, tool outputs, system constraints, and other context available to a fixed model with parameters θ . Let a_t denote externalized shared state such as notes, provenance records, code, diagrams, or a structured archive. Let e_t denote the task/environment state. Finally, let π denote the interface protocol: turn-taking rules, memory exposure, provenance requirements, tool permissions, disagreement mechanisms, summarization policy, and any other rule that governs coupling.

The joint state is

$$S_t = (h_t, c_t, a_t, e_t).$$

A minimal history-dependent model is

$$h_{t+1} = F_H(h_t, y_t, e_t, a_t; \pi)$$

$$y_t = G_\theta(c_t, x_t, a_t, e_t; \pi)$$

$$c_{t+1} = F_C(c_t, x_t, y_t, a_t; \pi)$$

$$a_{t+1} = F_A(a_t, x_t, y_t, r_t; \pi),$$

where x_t is the human contribution, y_t is the model contribution, and r_t denotes externally verifiable results such as experiments, calculations, test outcomes, citations, or measurements.

The full interaction is therefore a trajectory

$$\tau_{0:T}^{\pi} = \{S_0, S_1, \dots, S_T\}.$$

The central proposal is that many relevant properties of hybrid intelligence are functions of τ and π , not merely of h_0 and θ . Changing the protocol can alter the sequence of hypotheses considered, the errors detected, the information retained, the degree of human learning, and the final solution—even when the nominal human, model, and task remain unchanged.

Definition (Interface Intelligence). Interface intelligence is the set of empirically measurable, task-relevant system properties whose explanatory or predictive value depends on the protocol-mediated interaction trajectory and cannot be adequately recovered from isolated human and model performance alone.

This definition is intentionally conservative. It requires no claim about phenomenology, consciousness, or numerical identity. Emergence in this paper means only that a system-level variable or performance property cannot be adequately predicted from component measurements without modeling their organized interaction.

4 The interface signature: a multidimensional measurement object

A single performance score is too weak to characterize a hybrid system. A human–AI team can perform well while teaching the human nothing, encouraging over-reliance, destroying provenance, or becoming brittle under model substitution. Conversely, a collaboration may produce only modest immediate gains while substantially improving the human’s later capability. We therefore propose an interface signature rather than a scalar “hybrid intelligence quotient.”

Let the interface signature under protocol π be

$$\mathbf{I}(\pi) = [C_{\text{perf}}, C_{\text{info}}, B_{\text{coup}}, P_{\text{path}}, R_{\text{cap}}, O_{\text{opp}}, V_{\text{prov}}, T_{\text{swap}}].$$

Each component is separately measurable and should be reported as a vector. Collapsing these dimensions into a single score would require domain-specific value judgments and can conceal trade-offs.

4.1 Complementary task performance

Let P_H be human-only performance, P_M model-only performance, and P_{HM}^{π} performance of the coupled system under protocol π on a common task distribution. Define complementary performance gain

$$C_{\text{perf}}(\pi) = P_{HM}^{\pi} - \max(P_H, P_M).$$

$C_{\text{perf}} > 0$ is the strict performance criterion for complementary team performance. This matches the strongest interpretation of human–AI synergy in the empirical literature [4,5]. Importantly, augmentation of a weak human by a strong model is not, by itself, evidence of complementarity; the coupled system must exceed the stronger isolated component.

4.2 Informational synergy

Performance complementarity says that the joint system did better, but not how information was organized. Partial Information Decomposition (PID) provides a candidate formal language for separating redundant, unique, and synergistic information supplied by multiple sources about a target variable [10,11]. For appropriately defined human contribution H , model contribution M , and target Y ,

$$I(H, M; Y) = \text{Red} + \text{Un}_H + \text{Un}_M + \text{Syn}.$$

$\text{Syn}(H, M; Y)$ quantifies information about Y that is available only from the joint observation of H and M , under the chosen PID definition.

This is attractive for hybrid intelligence because “the whole contains task-relevant information unavailable in either part alone” is precisely the intuitive claim that often hides behind the word synergy. But PID is not a solved measurement problem: different decompositions embody different axioms, and synergy estimates can depend on estimator choice [10,11]. Any empirical use must preregister the decomposition, estimator, discretization or density model, and robustness checks. PID should be treated as one diagnostic, not as an oracle for emergence.

4.3 Bidirectional predictive coupling

A sustained collaboration should exhibit directed dependence in both directions: human behavior changes conditional on model outputs, and subsequent model behavior changes conditional on human inputs and accumulated context. Transfer entropy provides one possible statistic for time-asymmetric predictive dependence [12]. A conceptual pair is

$$TE_{H \rightarrow M} = I(H_{<t}; M_t | M_{<t})$$

$$TE_{M \rightarrow H} = I(M_{<t}; H_t | H_{<t}).$$

However, transfer entropy must not be casually renamed “causal information flow.” James, Barnett, and Crutchfield show that transfer-entropy-like quantities can misrepresent actual influence and can miss polyadic dependencies [13]. For causal claims, protocol variables or message availability should be experimentally randomized. We therefore use information-dynamic measures as descriptive coupling diagnostics and randomized intervention as the primary basis for causal inference.

4.4 Path dependence and trajectory integrity

If interface intelligence is a trajectory property, order and history should matter. Let π_1 and π_2 be two interaction protocols applied to the same task class with matched human/model capability. Define a path-dependence effect as the expected divergence in endpoint behavior or solution quality induced by protocol:

$$P_{\text{path}}(\pi_1, \pi_2) = \mathbb{E}[d(O_T^{\pi_1}, O_T^{\pi_2})],$$

where d is a preregistered outcome distance appropriate to the task.

The strongest tests use randomized protocols and tasks with known ground truth. For example, one condition may require the human to state an independent hypothesis before seeing the model; another may expose the model's answer first. Both use the same human and model class, but they alter the causal ordering of representations. If endpoint quality, calibration, or hypothesis diversity changes reliably, the trajectory—not merely component capability—is explanatory.

Trajectory integrity adds a second requirement: the final result should remain reconstructable from the transformations that produced it. In practical systems this means preserving enough provenance to distinguish an actual revision path from a retrospectively smoothed narrative.

4.5 Capability residue, debt, and mirage

Immediate joint performance does not tell us whether collaboration develops or erodes human capability. Let P_H^{pre} denote human-only performance before sustained collaboration, P_{HM}^{π} denote collaborative performance, and P_H^{post} denote the same human's performance on matched transfer tasks after the AI is removed.

When $P_{HM}^{\pi} > P_H^{\text{pre}}$, define capability residue

$$R_{\text{cap}} = \frac{P_H^{\text{post}} - P_H^{\text{pre}}}{P_{HM}^{\pi} - P_H^{\text{pre}}}.$$

$R_{\text{cap}} \approx 1$ indicates that most of the collaborative uplift became independently usable by the human. $R_{\text{cap}} \approx 0$ indicates that the gain was almost entirely scaffold-dependent. $R_{\text{cap}} < 0$ indicates deterioration relative to baseline; $R_{\text{cap}} > 1$ is possible if collaboration catalyzes learning that later exceeds the original joint uplift.

A simple capability-debt gap is

$$D_{\text{cap}} = P_{HM}^{\pi} - P_H^{\text{post}}.$$

High collaborative performance with high D_{cap} is a capability mirage: the system produces impressive outputs while leaving little transferable competence in the human. This construct should be interpreted only alongside task difficulty, practice effects, and a no-AI control group.

4.6 Opposition capacity

A sovereign hybrid system should not merely make the human more willing to accept machine output. It should preserve or improve the human's ability to reject the machine when rejection is warranted. We call this opposition capacity.

Operationally, a benchmark can inject plausible but controlled model errors on a subset of trials. O_{opp} is then measured through error detection, appropriate override, correction quality, and confidence calibration. The goal is not maximal disagreement. Blind contrarianism is as defective as blind compliance. The target is selective resistance: disagreement that tracks actual model failure.

This connects to trust-calibration research, which shows that over-trust and under-trust can both degrade human–AI performance and that calibrated reliance is an active design problem [14].

4.7 Provenance fidelity and transfer across model substitution

Relational capability becomes scientifically interesting when it can be separated from vendor- or model-specific dependence. We therefore propose two additional measurements.

Provenance fidelity V_{prov} measures whether participants can reconstruct why a claim or decision entered the shared state: which evidence, model output, human hypothesis, tool result, or correction produced it. This can be evaluated against an instrumented event log rather than by self-report alone.

Model-swap transfer T_{swap} measures how much capability survives when the AI component is replaced while the human and shared archive remain. If a human–model pair achieves high performance only with one specific model and collapses under substitution, much of the capability is relationally specific. If performance largely survives, the collaboration has produced portable structure. Model swap therefore provides a direct experimental probe of where capability lives.

Symbol	Construct	Operational test	Failure exposed
C_{perf}	Strict complementarity	Team vs. max(human, model)	Augmentation mistaken for synergy
C_{info}	Informational synergy	Preregistered PID on trial variables	Joint performance with no unique joint information
B_{coup}	Bidirectional coupling	Predictive dependence plus randomized perturbation	One-way automation masquerading as collaboration
P_{path}	Path dependence	Randomized order/protocol intervention	Endpoint-only analysis
R_{cap}	Capability residue	Human transfer performance after AI removal	Capability mirage / debt
O_{opp}	Opposition capacity	Seeded-error detection and correction	Over-trust / passive compliance
V_{prov}	Provenance fidelity	Reconstruction vs. instrumented event log	Retrospective smoothing / unauditability success
T_{swap}	Model-swap transfer	Replace model, preserve human plus archive	Vendor/persona-specific relational overfit

Table 1: Proposed interface-signature measurements. The construct is intentionally vector-valued; no universal scalar aggregation is assumed.

5 Sovereignty as an operational vector, not a slogan

Hybrid intelligence is often discussed in terms of augmentation, responsibility, or human control, but these terms become useful only when they can be operationalized. We propose a sovereignty vector

$$\Sigma = [E_{\text{exit}}, O_{\text{opp}}, T_{\text{swap}}, V_{\text{prov}}, A_{\text{goal}}],$$

where E_{exit} is exit capacity after AI withdrawal, O_{opp} is opposition capacity, T_{swap} is portability under model substitution, V_{prov} is provenance fidelity, and A_{goal} records whether final goals and irreversible actions remain under authorized human control.

The vector should not be automatically collapsed into a scalar. Different domains legitimately assign different importance to exit, speed, auditability, or performance. In a scientific-assistance system, provenance may dominate; in a safety-critical operational system, authorization and calibrated override may dominate.

This formalization also separates sovereignty from independence. A human may rationally depend on an AI system while remaining sovereign if the dependency is legible, contestable, portable, and subject to authorized exit. The relevant failure is captured dependency, not collaboration itself.

6 A falsifiable six-arm experimental program

The framework should stand or fall through intervention. We therefore propose a preregistered, longitudinal experiment that holds model class, task distribution, and participant pool fixed while manipulating the structure of coupling.

- **Arm A** — Human-only. Participants solve tasks without model assistance.
- **Arm B** — Model-only. The model solves the same task distribution under a fixed elicitation protocol.
- **Arm C** — One-shot consultation. The human receives one model response and then commits an answer.
- **Arm D** — Sustained dialogue. Human and model may iteratively question, revise, challenge, and test hypotheses for a fixed interaction budget.
- **Arm E** — Sustained dialogue + structured shared archive. Same as D, but with a provenance ledger that separates hypotheses, evidence, falsifications, unresolved questions, source links, and explicit revisions.
- **Arm F** — Model-swapped continuation. Participants begin under E with model M1; at a preregistered transition point the model is replaced by M2 while the shared archive and human continue. This directly tests portability and relational specificity.

6.1 Physics-facing benchmark: mechanistic discovery under controlled ground truth

A useful benchmark for a physics-oriented audience should reward scientific reasoning rather than rhetorical fluency. We propose families of synthetic but nontrivial mechanistic inference tasks in which the generating process is known to the experimenters but hidden from participants.

Candidate task families include:

- Parameter and structure inference for coupled ordinary differential equations.

- Identification of latent coupling topology in oscillator networks such as Kuramoto-type systems.
- Distinguishing stochastic from deterministic mechanisms under finite noisy observations.
- Recovering conserved quantities or approximate invariants from generated trajectories.
- Symbolic regression of compact governing equations from simulated data.
- Discriminating rival mechanistic models that fit the training observations but diverge under intervention.
- Debugging intentionally flawed derivations where the AI occasionally receives or produces a plausible but incorrect intermediate step.

These tasks have four advantages. First, they provide objective ground truth. Second, they admit genuine division of labor: humans may contribute physical intuition and hypothesis restructuring while models contribute search, algebra, coding, and breadth. Third, they make error detection measurable. Fourth, they allow controlled perturbations of information availability and interaction order.

6.2 Longitudinal structure

Each participant should complete multiple sessions over time rather than a single benchmark exposure. Session 1 estimates baseline human capability. Sessions 2–4 create the collaborative trajectory. Session 5 removes the AI to measure capability residue on transfer tasks. A later session reintroduces a different model to measure T_{swap} .

The shared archive in Arm E/F should be externalized rather than hidden in model memory. This makes the information state inspectable, reproducible, and portable across models. It also allows a direct comparison between “same model, different archive” and “different model, same archive.”

6.3 Primary and secondary endpoints

Primary endpoint:

C_{perf} — strict complementary performance relative to the better isolated component.

Secondary endpoints:

- Calibration error and Brier score where probabilistic predictions are possible.
- Hypothesis diversity before convergence.
- Counterexample generation rate.
- Time-to-correct first false hypothesis.
- Appropriate override rate under seeded model errors.
- Provenance fidelity against event logs.
- Capability residue after AI withdrawal.
- Model-swap transfer.
- Interaction efficiency: performance per token, turn, wall-clock minute, and tool call.
- Information-theoretic diagnostics, including preregistered PID estimates where the response representation supports them.

6.4 Statistical design

The primary analysis should use a hierarchical model with participant, task instance, and model identity as random effects, and protocol arm as a fixed experimental factor. The model should explicitly estimate heterogeneity rather than report only a grand mean, because existing evidence shows that human–AI synergy varies strongly by task type [4].

For the core complementarity endpoint, the null is $C_{\text{perf}} \leq 0$. For learning, the null is $P_H^{\text{post}} - P_H^{\text{pre}} \leq$ the matched no-AI practice effect. For model-swap transfer, the relevant estimand is the degradation or retention relative to the pre-swap collaborative baseline. Confidence or credible intervals should be reported for all effects; multiplicity correction should be preregistered for confirmatory secondary hypotheses.

Information-dynamic analyses require substantially more samples than ordinary accuracy comparisons. PID and transfer-entropy estimates should therefore be treated as confirmatory only where sample size and estimator validation are adequate; otherwise they should remain exploratory.

7 Physics interpretation: coupling architecture rather than metaphorical thermodynamics

Hybrid intelligence invites physical analogies, but analogy can easily outrun measurement. The useful import from physics is not a claim that human–AI dialogue literally obeys a new thermodynamic law. It is the discipline of studying coupled systems through state variables, interaction terms, perturbations, response functions, path dependence, stability, and scale.

Under this view, the interface protocol π functions as a coupling operator. It determines which variables are exposed, which transformations are allowed, the latency and bandwidth of exchange, the persistence of state, and the feedback available to each subsystem. Two systems with identical components can exhibit different macroscopic behavior when the coupling topology changes. Human–AI systems should be tested with the same seriousness.

This perspective suggests several measurable quantities:

1. Coupling strength: sensitivity of each subsystem’s next-state distribution to perturbations in the other.
2. Asymmetry: whether influence is primarily model→human, human→model/context, or bidirectional.
3. Memory depth: how far into the interaction history one must condition before prediction saturates.
4. Hysteresis/path dependence: whether returning to the same nominal information state after a different history yields different behavior.
5. Robustness: how output degrades under message removal, archive corruption, model substitution, or delayed feedback.
6. Higher-order synergy: whether joint variables explain target information unavailable in dyadic marginals.

Transfer entropy and information decomposition provide relevant mathematical tools, but neither should be overinterpreted. Schreiber’s transfer entropy was designed to

quantify time-asymmetric predictive dependence [12], while later work demonstrates that such statistics need not equal causal flow and may miss higher-order organization [13]. Likewise, information decomposition makes synergy mathematically explicit, but no single PID measure has universal status [10,11]. The correct physics instinct is therefore plural measurement plus intervention, not a single seductive scalar.

A genuine “phase transition” claim would require evidence of qualitative regime change, scaling behavior, or order-parameter structure as coupling variables are varied. Until such evidence exists, phase-transition language should remain metaphorical. The framework instead asks a simpler question: do measurable system-level properties change non-linearly as the interaction protocol changes, and are those changes reproducible?

8 Testable hypotheses

H1 — Protocol dependence. With human and model capability controlled, sustained dialogue and provenance-structured interaction will explain significant variance in performance beyond one-shot consultation.

H2 — Complementarity is conditional, not generic. Strict $C_{\text{perf}} > 0$ will occur primarily on task classes with exploitable information or capability asymmetry, consistent with existing complementarity theory [5], rather than as a universal effect of “more interaction.”

H3 — Performance and learning dissociate. Some protocols will maximize immediate P_{HM} while producing low R_{cap} , demonstrating capability mirage; other protocols will sacrifice some immediate speed while increasing post-withdrawal human performance.

H4 — Opposition capacity predicts robust synergy. Participants who appropriately detect seeded model errors will show better calibration and higher performance under distribution shift than participants with high raw reliance.

H5 — Provenance increases portability. Structured archives will reduce performance loss under model substitution and improve reconstruction of decision lineage, even if they impose a short-term interaction cost.

H6 — Relational capability exists. Model-swap and tool-withdrawal interventions will reveal capabilities that are neither stable properties of the human nor the model alone, but specific to the learned interaction configuration.

H7 — Coupling magnitude is not sufficient. High bidirectional predictive dependence will not necessarily imply high-quality hybrid intelligence; some tightly coupled systems will exhibit synchronized error, anchoring, or mutual reinforcement. Coupling must therefore be interpreted jointly with correctness, calibration, opposition, and provenance.

9 Failure modes the framework must detect

A research framework is useful only if it can diagnose its own attractive failures.

Automation capture. The human becomes a confirmation layer around a dominant model. Team output may be strong, but O_{opp} and R_{cap} collapse.

Mutual error amplification. Iteration increases confidence and coherence while both parties orbit an early false assumption. This is a tightly coupled failure, not a lack of coupling.

Provenance collapse. The final answer is good but no participant can reconstruct which evidence, calculation, or correction justified it. This makes the capability difficult to audit or transfer.

Archive ossification. External memory becomes a source of anchoring: obsolete hypotheses persist because they are repeatedly retrieved. Provenance systems therefore need revision state, not only retention.

Model-persona overfitting. A human becomes exceptionally effective with one model's conversational style but cannot transfer the capability. This should appear as low T_{swap} despite high P_{HM} .

False emergence. The team exceeds the average human and average model but not the stronger isolated component. This is augmentation, not strict complementarity.

Metric theater. A complicated information-theoretic statistic is treated as proof of cognition, consciousness, or causality. The framework explicitly rejects this inference.

10 Falsifiability: what would make interface intelligence unnecessary?

The proposal is not protected from failure. It would lose much of its scientific value if, after controlling for component capability and task difficulty, interaction protocol and trajectory variables add negligible predictive power. Specifically, the framework would be weakened if:

- one-shot and sustained protocols produce equivalent outcomes once token/compute budgets are matched;
- structured provenance never improves error correction, transfer, or auditability;
- post-withdrawal human performance is fully explained by generic practice effects;
- model swaps do not expose relationally specific capability;
- path-order interventions fail to alter outcomes;
- interface-signature variables do not predict robustness beyond ordinary component benchmarks.

In that case, “interface intelligence” would be unnecessary vocabulary for effects better explained by model quality, human expertise, or resource allocation. That is an acceptable outcome. The concept earns survival only by explaining and predicting phenomena that simpler decompositions miss.

11 Research implications

The framework shifts several common questions.

Instead of asking only “How intelligent is the model?”, ask “What capability appears under which coupling architecture?”

Instead of asking only “Does AI improve the human?”, ask “Which parts of the gain remain after the AI disappears?”

Instead of asking only “Does the human trust the AI?”, ask “Does reliance track actual

competence, and can the human still oppose the system when it fails?”

Instead of asking only “Who produced the answer?”, ask “Which information was unique, redundant, or synergistic across the trajectory?”

Instead of treating memory as convenience, treat shared memory as an experimental variable that can create both capability and anchoring.

Instead of assuming a model upgrade monotonically improves the team, test whether replacing the model destroys learned relational structure.

These questions matter for more than productivity. Long-duration scientific work, education, medicine, engineering, governance, and autonomous systems increasingly involve repeated interaction rather than isolated queries. In such settings, the object being designed is not merely a model output. It is a trajectory of coupled adaptation.

12 Relation to consciousness claims

This framework is intentionally agnostic about AI consciousness. A coupled human–AI system can exhibit emergent task properties, synergistic information, history dependence, and relational capability without those facts establishing a new conscious subject. System-level cognition, system-level computation, and system-level consciousness are different propositions and require different evidence.

The appropriate epistemic rule is therefore strict: no transition from interaction-level emergence to phenomenological consciousness is licensed by the present formalism. If future work develops validated measures of consciousness applicable across substrates, those measures could be studied alongside the interface signature. They should not be smuggled into it.

13 Conclusion

Hybrid intelligence will not be understood by benchmarking humans and models separately and assuming that their combination is the sum of two capability scores. The evidence already shows that human–AI combinations can underperform their strongest component [4]. The scientific challenge is therefore not merely to combine intelligence, but to understand the organization that makes combination productive, developmental, robust, and governable.

Interface intelligence names that organization as an empirical target. The proposal is that human–AI capability can be distributed across a trajectory: human state, model context, shared archive, interaction protocol, tools, corrections, and accumulated history. The resulting system may contain capabilities that vanish when the relationship is broken, transfer when its structure is externalized, or become harmful when coupling outruns calibration.

The strongest version of the research program is therefore experimental. Manipulate the coupling. Preserve the components. Measure strict complementarity. Withdraw the scaffold. Swap the model. Seed errors. Trace provenance. Quantify information where the data support it. Ask not only whether the system performs, but where the capability lives and what survives when the configuration changes.

If the interaction trajectory contributes no independent explanatory power, the concept should be discarded. If it does, then a major unit of intelligence research has been hiding in plain sight: not the human, not the model, but the structured path through which they change one another.

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Research disclosure

This working paper was developed through iterative human–AI dialogue and literature synthesis. The named constructs introduced here are hypotheses and proposed measurement objects unless explicitly attributed to prior literature. Empirical claims are limited to the cited sources. The framework does not claim or assume AI consciousness.

Governing epistemic rule

No question is forbidden by strangeness; no answer is promoted beyond its evidence.